3. Domain: Digital India RISC-V (DIR-V)

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	VEGA Series of Processors ASTRA portfolio of System/Peripheral Ips
	Name of the Project under which technology is developed	Microprocessor Development Programme
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 9
	Brief Write-up on the Technology Developed	The VEGA processors comprise of a series of 32-bit/64-bit Single/Multi-core Superscalar In-order/Out-of-Order high performance processors based on RISC-V ISA integrated with Multi-level Caches, MMU and Coherent Interconnect. The ASTRA portfolio comprises of a wide range of System and Peripheral Ips which are Silicon proven, comprising of the robust RTL, extensively verified and fully synthesizable technology independent IP cores which form the building blocks for an SoC implementation
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Keith Fernandes, C-DAC Thiruvanathapuram 9895183437 keith@cdac.in
2	**Brief details of Technology Developed**	• RISC-V ISA enabled Floating Point Unit (FPU) • RISC-V ISA enabled Posit Arithmetic Unit • Level-1 Instruction and Data cache controller • Static RAM controller IP core
	Name of the Project under which technology is developed	National Micro-processor Initiative
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL-6
	Brief Write-up on the Technology Developed	**RISC-V ISA Enabled Floating Point Unit (FPU)** The RISC-V ISA–enabled Floating Point Unit (FPU) implements IEEE-754 compliant single- and double-precision floating-point operations in alignment with the RISC-V F and D extensions. The unit supports arithmetic, comparison, and conversion instructions, and is designed for seamless integration with RISC-V processor pipelines. It is optimized for high performance and low power, making it suitable for HPC, AI acceleration, and scientific computing workloads. **RISC-V ISA Enabled Posit Arithmetic Unit** The RISC-V ISA–enabled Posit Arithmetic Unit provides hardware support for posit number representation as an alternative to IEEE-754 floating point. By enabling dynamic precision and improved accuracy for low-bit-width

Sr. No.	Parameters	Brief details
		computations, the posit unit enhances energy efficiency and numerical robustness. Integrated via custom or extended RISC-V instructions, this unit is particularly beneficial for AI/ML inference, signal processing, and edge computing applications. **Level-1 Instruction and Data Cache Controller** The Level-1 Instruction and Data Cache Controller implements low-latency, high-bandwidth cache management for RISC-V processors. It supports configurable cache sizes, associativity, and line sizes, with features such as cache coherency hooks, write-back/write-through policies, and replacement algorithms. The controller significantly improves processor performance by reducing memory access latency and optimizing instruction fetch and data access patterns. **Static RAM (SRAM) Controller IP Core** The Static RAM Controller IP Core provides a reliable and configurable interface between processor subsystems and on-chip or off-chip SRAM. It supports parameterizable data widths, address widths, and access timing, along with features such as byte enables, burst access, and error detection options. Designed for low-latency memory access, the controller is suitable for embedded systems, SoCs, and security-sensitive applications.
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